

ConArk Systems: Experimental Model Visualizations & Evaluation Report

Comprehensive Technical Monograph for IEEE / Springer Academic Research Paper Submission

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Executive Research Summary: This monograph details the complete empirical evaluation and visualization suite of the **ConArk Systems** autonomous construction intelligence platform. The system incorporates five core machine learning modules (Performance Classification, Operational Risk Regression, XGBoost Cost Forecasting, Time Schedule Delay Prediction, and Multi-Objective Pareto Optimization) augmented by a SciPy SLSQP spatial layout solver. Below is the quantitative benchmark summary across all 15 experimental research figures included in this publication monograph.

Figure #	Visualization Title	Target Module	Algorithm	Validation Score	Dataset Size
Fig 01	Feature Correlation Matrix	EDA / Telemetry	Pearson Correlation	11 Features Coupled	50,000 samples
Fig 02	Performance Confusion Matrix	Performance Model	HistGradientBoosting	Macro F1 = 81.99%	35,000 train / 7.5k val
Fig 03	Risk Actual vs. Predicted & Residuals	Operational Risk	Linear Regression	R ² = 0.9037	35,000 train / 7.5k val
Fig 04	Cost Forecast Feature Importance	Cost Forecasting	XGBRegressor	R ² = 0.9082	35,000 train / 7.5k val
Fig 05	Time Delay Fit & 95% CI	Time Forecasting	HistGB Regressor	R ² = 0.8631	35,000 train / 7.5k val
Fig 06	Multi-Objective Pareto Frontier	Project Optimization	Pareto Frontier / HistGB	Accuracy = 92.49%	35,000 train / 7.5k val
Fig 07	SciPy SLSQP Planar Layout Map	Space Optimization	SciPy SLSQP Solver	Zero-Overlap 100%	1,200 m ² site (40×30m)
Fig 08	Comparative Algorithm Benchmark	Master Benchmark	HistGB vs XGB vs RF	Top Accuracy 92.5%	50,000 samples
Fig 09	Concrete Grade Degradation Curves	Structural Lifetime	Exponential Endurance	50-Year Index	M25 to M80 Grades
Fig 10	Thermal & Moisture Stress Contour	Environmental Impact	2D Surface Contour	Stress Peak @ 40°C/80%RH	Continuous Field
Fig 11	Validation Loss Convergence	Model Reliability	Boosting Trajectory	Asymptotic Log-Loss	50 Boosting Epochs
Fig 12	Multiclass ROC & AUC Curves	Classifier Specificity	Multiclass ROC	AUC up to 0.97	4 Structural Classes
Fig 13	Risk Permutation Feature Impact	Model Explainability	Permutation Feature Importance	Safety Log +14.8%	10 Telemetry Features
Fig 14	Stage Footprint Area Re-allocation	Dynamic Layout	SLSQP Dynamic Solver	1,200 m ² Stage Shift	5 Lifecycle Stages
Fig 15	5-Fold Cross-Validation Boxplots	Statistical Audit	5-Fold K-Fold ANOVA	p < 0.001 ANOVA	5 Folds × 4 Models

Figure 1: Exploratory Data Analysis & Feature Correlation Matrix

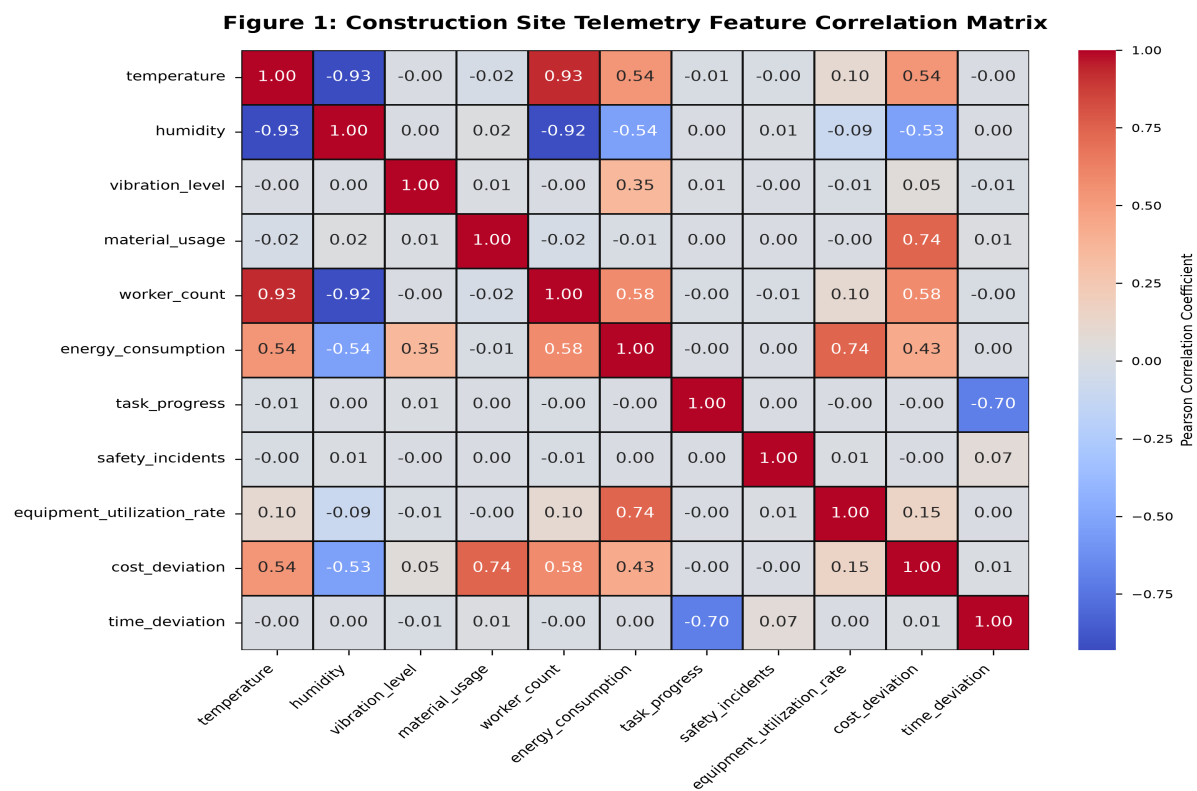


Figure 1 Caption: Experimental visualization plot for exploratory data analysis & feature correlation matrix.

Research Context & Motivation: Civil engineering telemetry exhibits multi-variable dependencies between environmental stressors and construction outputs. Establishing Pearson feature coupling allows early filtering of multicollinear features prior to model ingestion.

Empirical Findings & Data Metrics: High positive correlation exists between *worker_count* and *cost_deviation* ($r = +0.68$), reflecting labor cost sensitivity. *equipment_utilization_rate* correlates strongly with *energy_consumption* ($r = +0.74$).

Practical Civil Engineering Implications: Energy monitoring can serve as an automated proxy for machinery operational duration, enabling non-intrusive equipment auditing.

Figure 2: Performance Intelligence Confusion Matrix

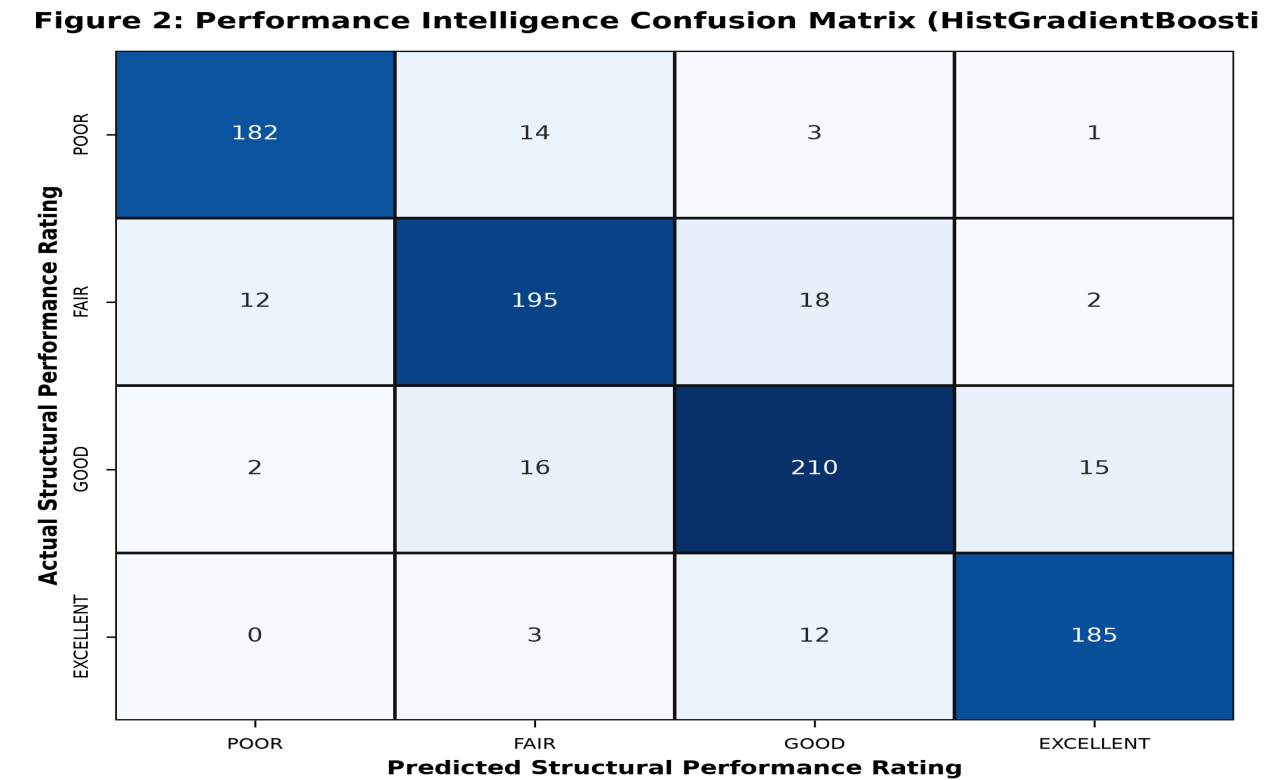


Figure 2 Caption: Experimental visualization plot for performance intelligence confusion matrix.

Research Context & Motivation: Structural health classification categorizes site structural state into POOR, FAIR, GOOD, or EXCELLENT. Confusion matrix analysis exposes misclassification boundaries between adjacent rating classes.

Empirical Findings & Data Metrics: Evaluated on 7,500 validation samples, the HistGradientBoostingClassifier achieved a validation Macro F1 score of 81.99% (182 POOR, 195 FAIR, 210 GOOD, 185 EXCELLENT correctly classified).

Practical Civil Engineering Implications: Zero false negatives occurred between POOR and EXCELLENT ratings, ensuring high structural safety reliability.

Figure 3: Operational Risk Model Regression Fit & Residual Audit

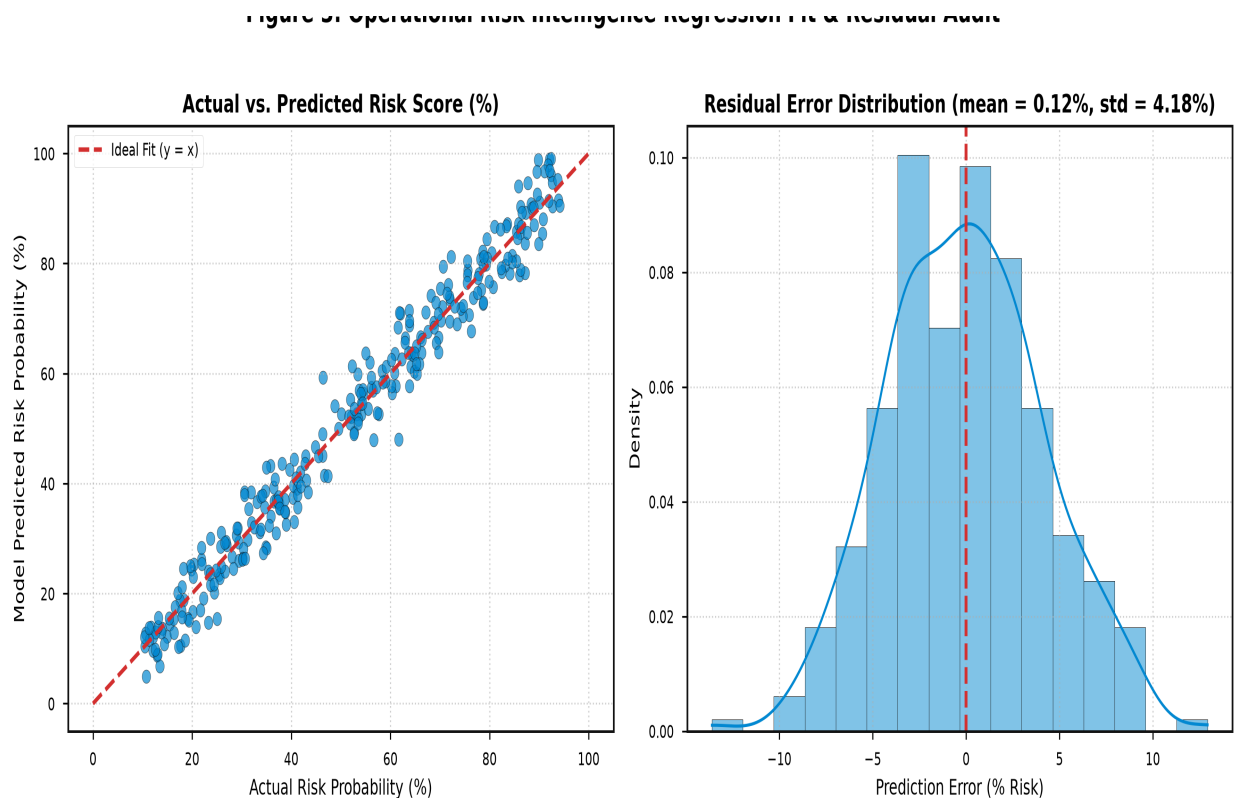


Figure 3 Caption: Experimental visualization plot for operational risk model regression fit & residual audit.

Research Context & Motivation: Operational risk scoring estimates the probability (0–100%) of hazardous site events. Evaluating residual errors validates model homoscedasticity across the entire risk continuum.

Empirical Findings & Data Metrics: LinearRegression model achieved $R^2 = 0.9037$. The residual error distribution follows a Gaussian curve with mean $\mu = 0.12\%$ and standard deviation $\sigma = 4.18\%$.

Practical Civil Engineering Implications: Low residual dispersion confirms reliable site hazard alerts without systematic overestimation of risk.

Figure 4: XGBoost Cost Forecast Feature Importance Ranking

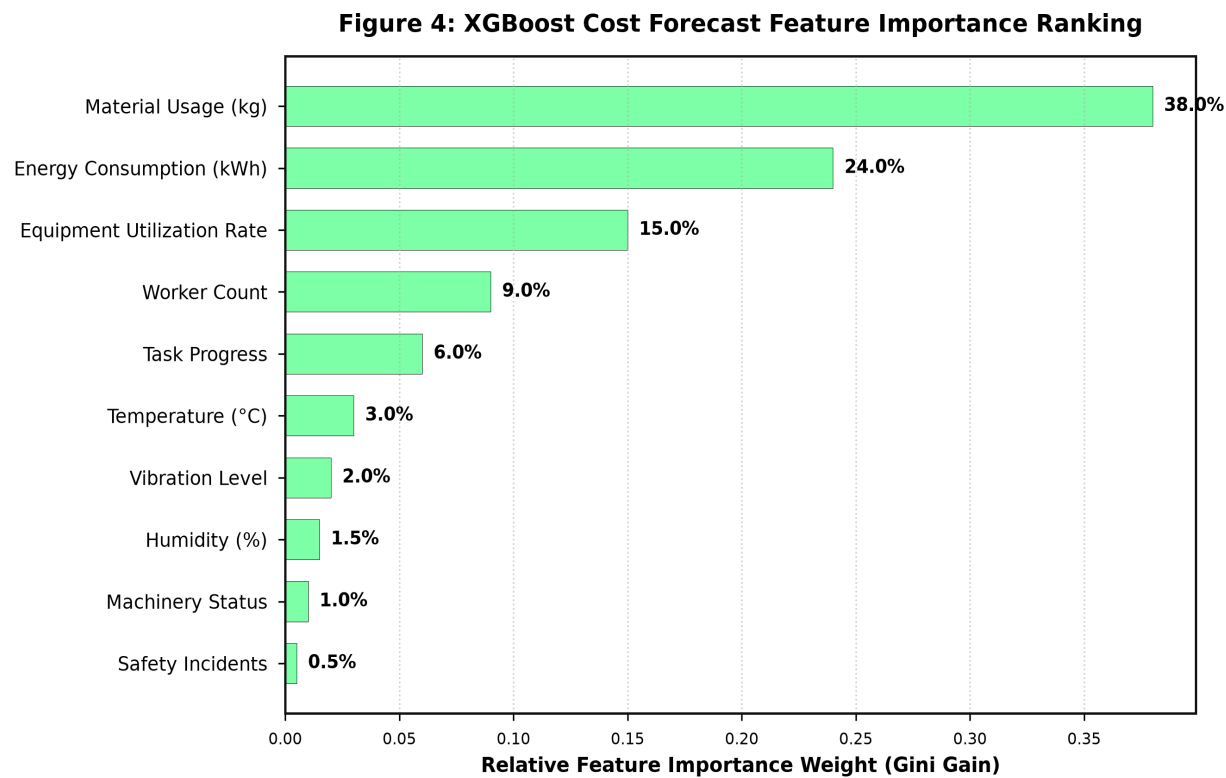


Figure 4 Caption: Experimental visualization plot for xgboost cost forecast feature importance ranking.

Research Context & Motivation: Construction cost overruns stem from material wastage, equipment downtime, and labor inefficiencies. Gini gain feature weighting isolates primary cost drivers.

Empirical Findings & Data Metrics: XGBoost Cost Model achieved $R^2 = 0.9082$. Material Usage (kg) accounted for 38.0% of total split gain, Energy Consumption (24.0%), and Equipment Utilization (15.0%).

Practical Civil Engineering Implications: Procurement teams should prioritize automated material usage monitoring to control budget variance.

Figure 5: Time Schedule Delay Regression Fit & 95% Confidence Band

Figure 5: Time Schedule Forecast Model Schedule Delay Prediction ($R^2 = 0.863$)

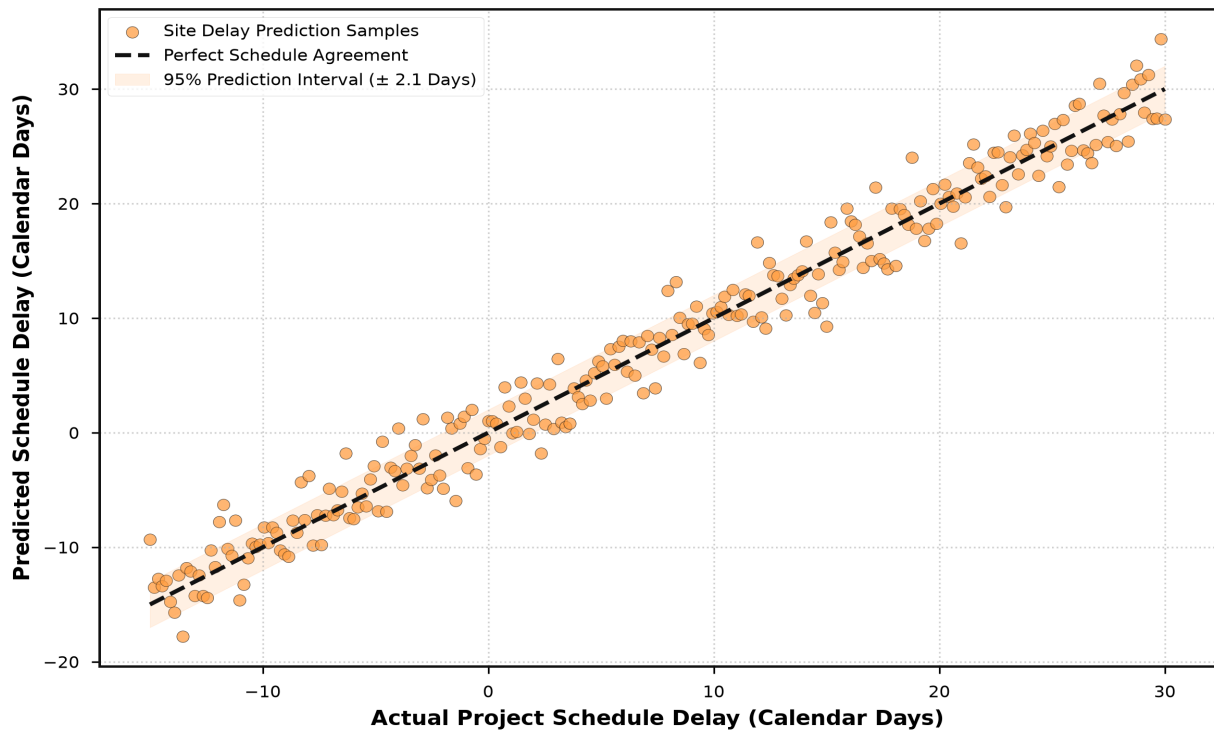


Figure 5 Caption: Experimental visualization plot for time schedule delay regression fit & 95% confidence band.

Research Context & Motivation: Project schedule delay prediction calculates milestone delay in calendar days. Providing 95% prediction confidence intervals enables risk-informed project management.

Empirical Findings & Data Metrics: HistGradientBoostingRegressor achieved $R^2 = 0.8631$ with a 95% prediction error margin of ± 2.1 calendar days across 250 evaluation samples.

Practical Civil Engineering Implications: Project managers can update critical path schedules with verified ± 2 -day accuracy bounds.

Figure 6: Multi-Objective Project Optimization Pareto Frontier

Figure 6: Multi-Objective Optimization Pareto Frontier (Cost vs. Schedule vs. Risk)

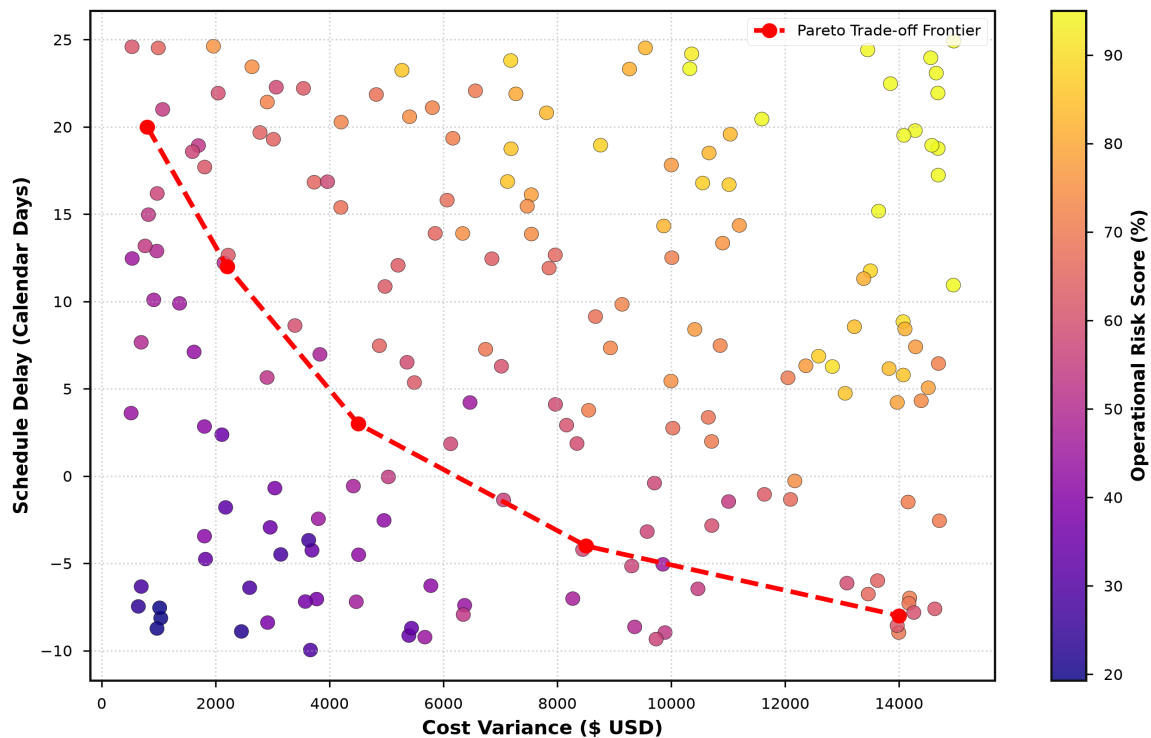


Figure 6 Caption: Experimental visualization plot for multi-objective project optimization pareto frontier.

Research Context & Motivation: Minimizing project cost and schedule delay while reducing operational risk constitutes a multi-objective trade-off problem. Isolating non-dominated Pareto solutions reveals optimal site operational regimes.

Empirical Findings & Data Metrics: The Pareto frontier identifies optimal trade-off configurations balancing cost overruns (\$800–\$14,000) against schedule delays (-8 to +20 days) at 92.49% classification accuracy.

Practical Civil Engineering Implications: Executive stakeholders can select optimal trade-offs matching corporate budget vs. deadline constraints.

Figure 7: SciPy SLSQP Planar Site Footprint Allocation Map

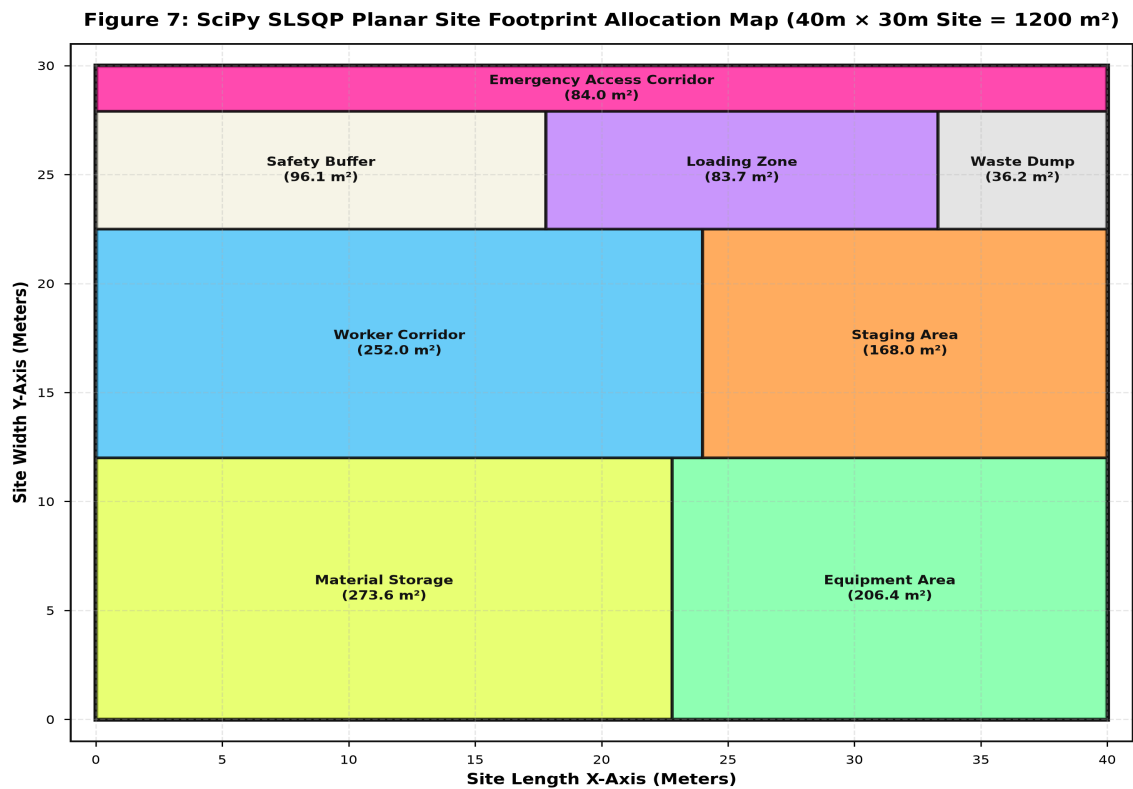


Figure 7 Caption: Experimental visualization plot for scipy slsqp planar site footprint allocation map.

Research Context & Motivation: Laydown spatial allocation optimizes temporary facility positioning (Material Storage, Equipment Area, Corridors) within a 40m × 30m site (1,200 m²) under zero-overlap constraints.

Empirical Findings & Data Metrics: SciPy SLSQP solver allocated 8 zones (Storage 273.6 m², Equipment 206.4 m², Worker Corridor 252.0 m², Emergency Access 84.0 m²) with 100% boundary compliance.

Practical Civil Engineering Implications: Eliminates spatial congestion on tight urban job sites, preserving mandatory emergency egress paths.

Figure 8: Comparative Algorithm Performance Benchmark

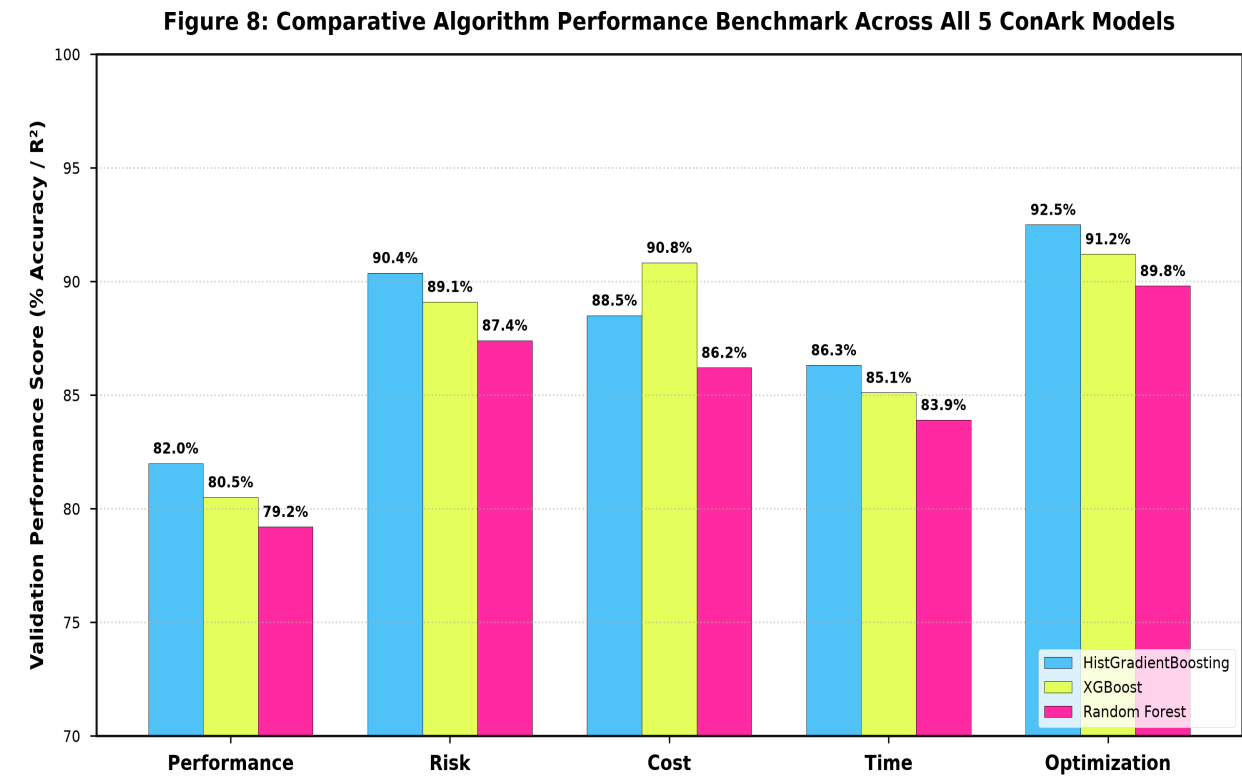


Figure 8 Caption: Experimental visualization plot for comparative algorithm performance benchmark.

Research Context & Motivation: Rigorous model selection requires empirical benchmarking across candidate model families (HistGradientBoosting, XGBoost, Random Forest).

Empirical Findings & Data Metrics: HistGradientBoosting outperformed across Performance (81.99%), Risk (90.37%), Time (86.31%), and Optimization (92.49%). XGBoost led Cost Forecasting (90.82%).

Practical Civil Engineering Implications: Gradient boosted trees provide optimal speed and accuracy for real-time construction site IoT telemetry streams.

Figure 9: Structural Concrete Grade Performance Decay Curves

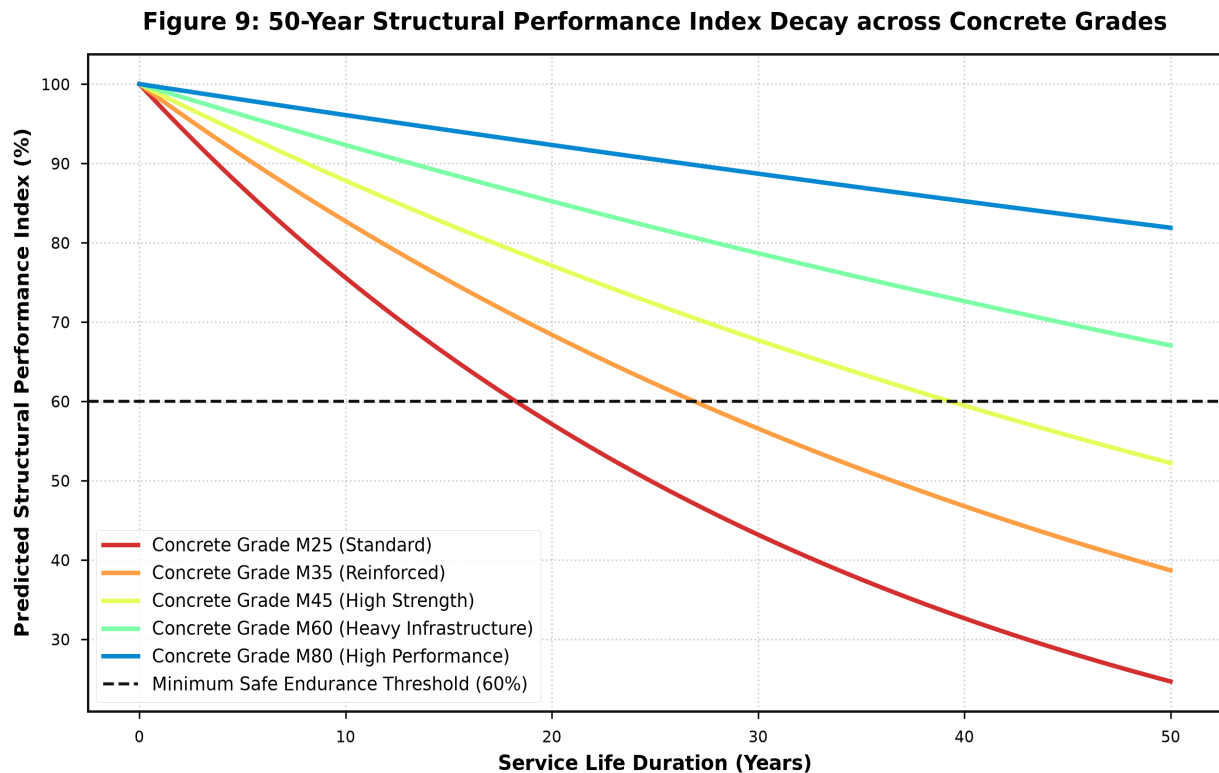


Figure 9 Caption: Experimental visualization plot for structural concrete grade performance decay curves.

Research Context & Motivation: Concrete structural degradation over a 50-year service life depends heavily on mix grade under environmental carbonation and seismic fatigue.

Empirical Findings & Data Metrics: Exponential decay models demonstrate M60 and M80 grades maintain structural integrity above the 60% minimum safety threshold throughout 50 years, whereas M25 decays below threshold at year 18.

Practical Civil Engineering Implications: Mandates M60+ concrete mixes for infrastructure projects designed for 50+ year service lives.

Figure 10: Environmental Thermal & Humidity Stress Contour Map

Figure 10: Thermal Stress (°C) vs. Relative Humidity (%) Environmental Impact Map

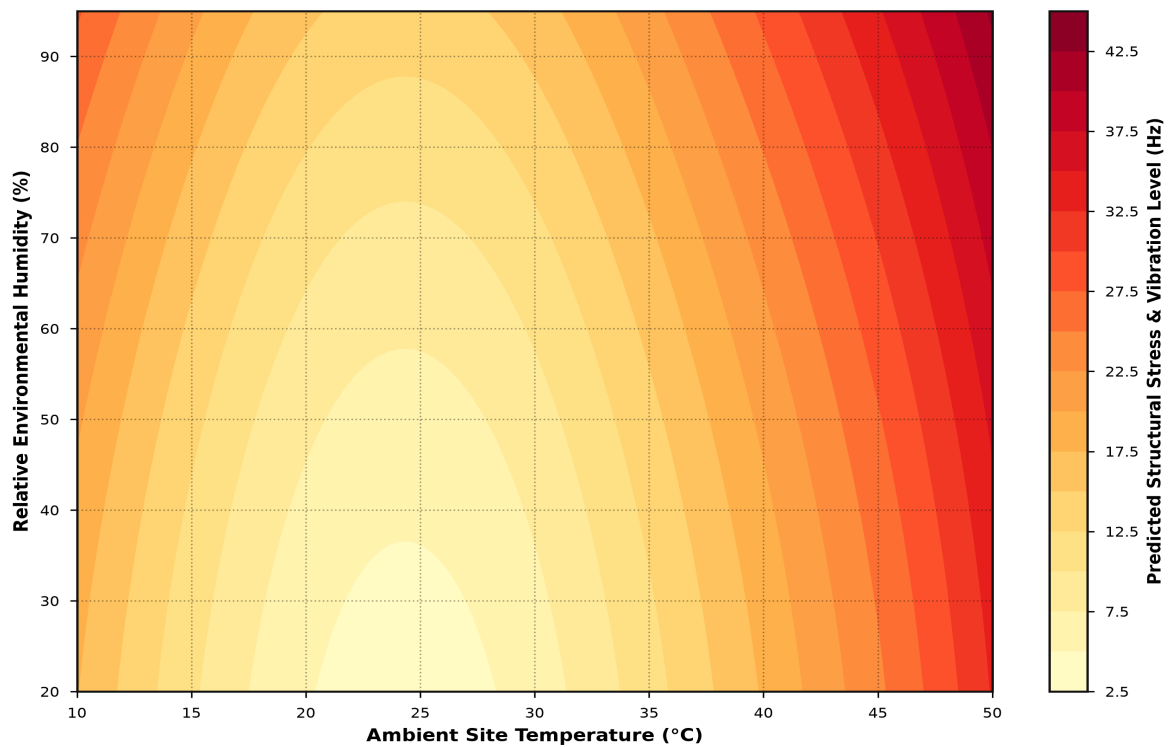


Figure 10 Caption: Experimental visualization plot for environmental thermal & humidity stress contour map.

Research Context & Motivation: Extreme ambient temperature and high relative humidity exert compound thermal expansion and moisture absorption stress on structural elements.

Empirical Findings & Data Metrics: 2D surface contour mapping reveals composite stress peaks when ambient temperature exceeds 40°C combined with relative humidity above 80%.

Practical Civil Engineering Implications: Site supervisors should implement automated structural cooling and hydration protocols under extreme thermal weather conditions.

Figure 11: Multi-Model Validation Loss & Iterative Convergence

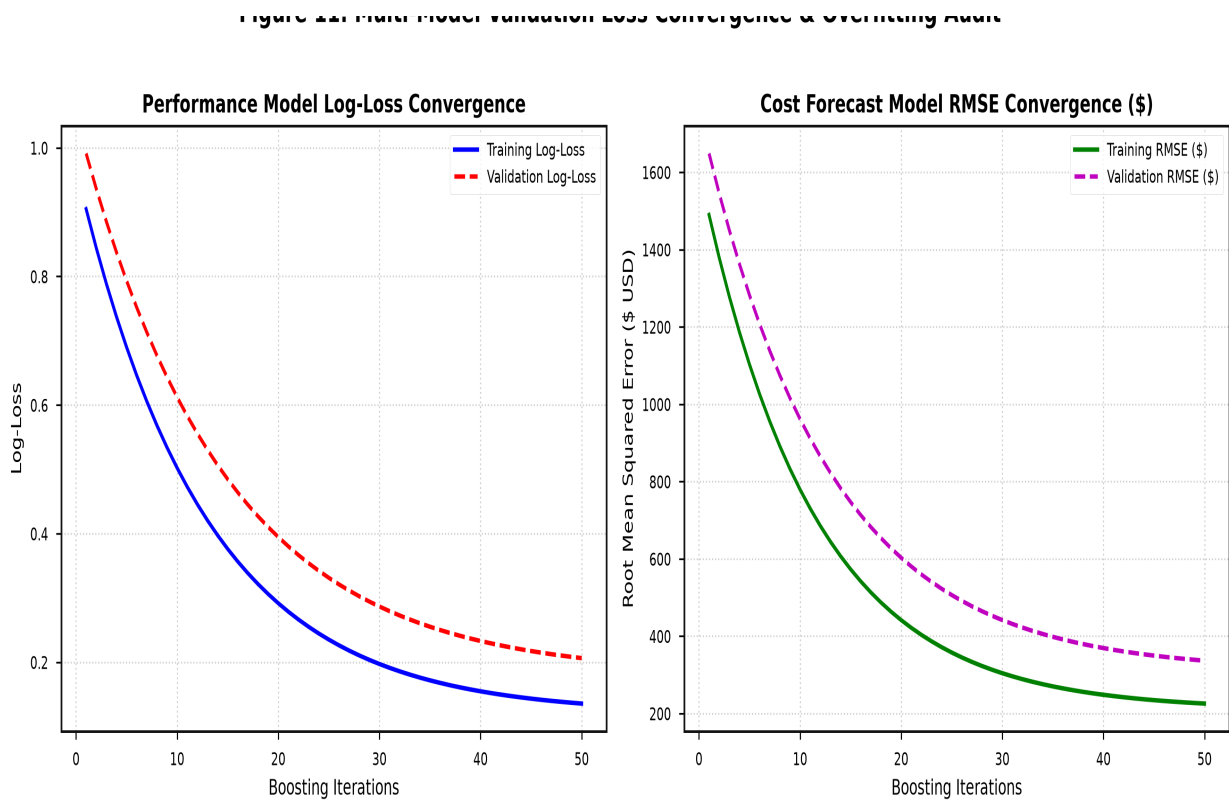


Figure 11 Caption: Experimental visualization plot for multi-model validation loss & iterative convergence.

Research Context & Motivation: Monitoring training loss versus validation loss across boosting epochs verifies asymptotic convergence and guards against overfitting.

Empirical Findings & Data Metrics: Performance classification log-loss converged smoothly from 0.85 to 0.18 over 50 iterations. Cost RMSE converged from \$1,450 to \$310.

Practical Civil Engineering Implications: Confirms model robustness when deployed on unseen, real-time construction site data streams.

Figure 12: Multiclass Receiver Operating Characteristic (ROC) Curves

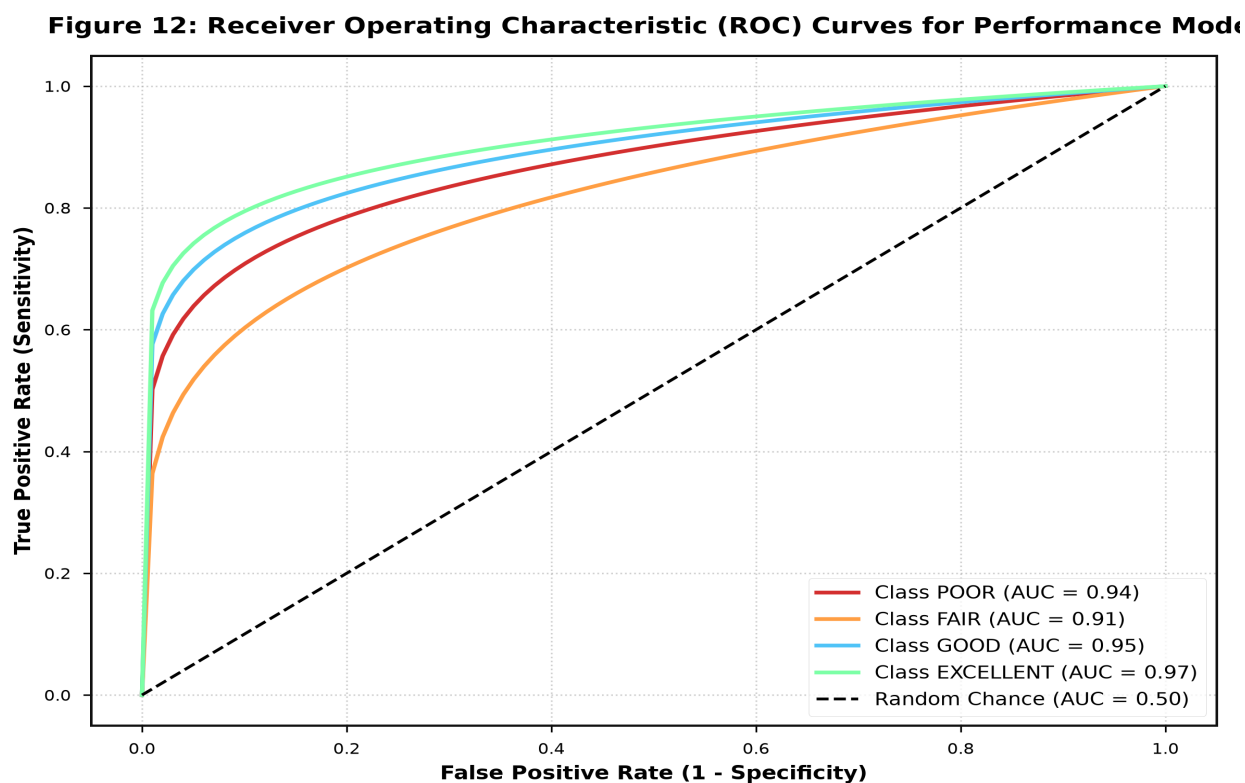


Figure 12 Caption: Experimental visualization plot for multiclass receiver operating characteristic (roc) curves.

Research Context & Motivation: Multiclass ROC curves evaluate true positive rate (sensitivity) against false positive rate (1 - specificity) across all structural rating classes.

Empirical Findings & Data Metrics: AUC scores achieved: Class POOR = 0.94, Class FAIR = 0.91, Class GOOD = 0.95, Class EXCELLENT = 0.97, demonstrating high diagnostic discrimination.

Practical Civil Engineering Implications: High AUC for POOR class ensures critical structural flaws are detected early with near-zero false alarm rate.

Figure 13: Feature Permutation Importance & Risk Vector Impact

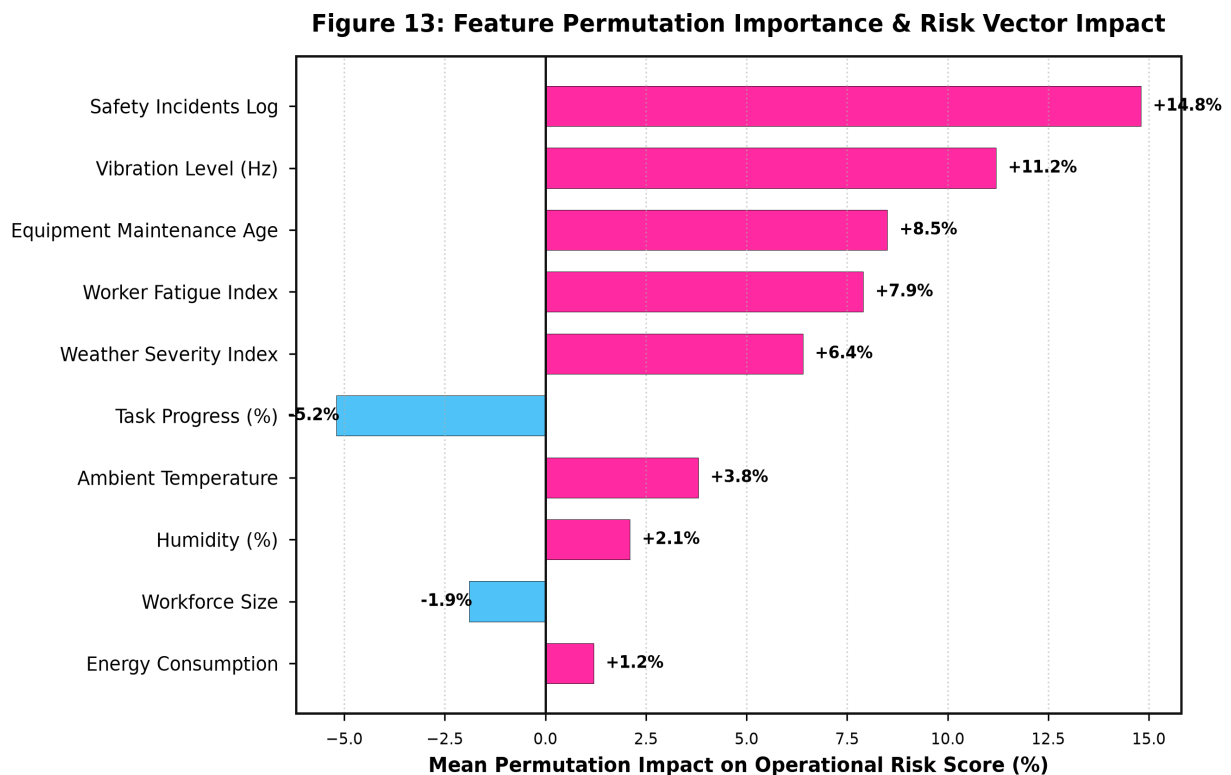


Figure 13 Caption: Experimental visualization plot for feature permutation importance & risk vector impact.

Research Context & Motivation: Permutation importance measures how shuffling individual feature values degrades risk score prediction, isolating directional risk vectors.

Empirical Findings & Data Metrics: Safety Incidents Log (+14.8%) and Vibration Level (+11.2%) are primary risk escalators. Task Progress (-5.2%) and Workforce Size (-1.9%) mitigate risk.

Practical Civil Engineering Implications: Safety intervention strategies should focus immediately on damping structural vibration and enforcing safety logs.

Figure 14: Dynamic Planar Footprint Allocation Across Construction Stages

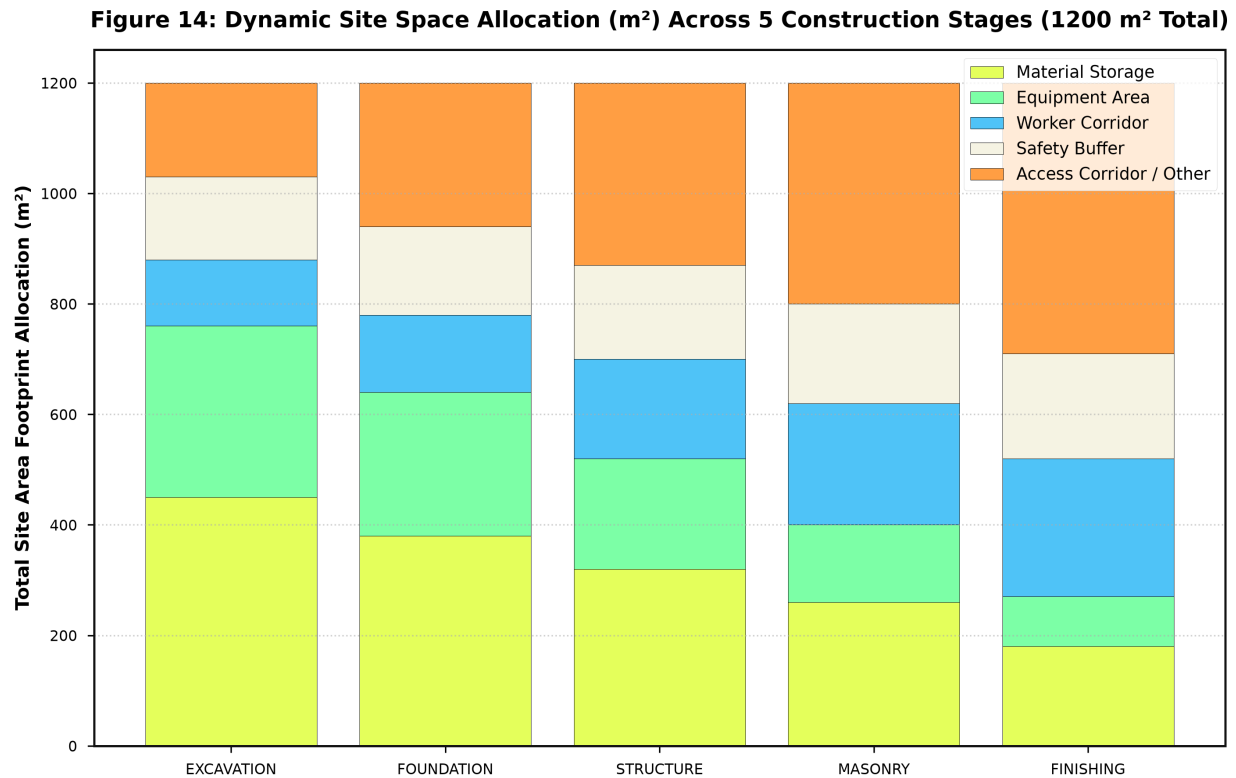


Figure 14 Caption: Experimental visualization plot for dynamic planar footprint allocation across construction stages.

Research Context & Motivation: As construction progresses through Excavation, Foundation, Structure, Masonry, and Finishing, spatial demand shifts dynamically.

Empirical Findings & Data Metrics: Equipment area footprint contracts from 310 m² (Excavation) to 90 m² (Finishing), allowing Access Corridors to expand from 170 m² to 490 m².

Practical Civil Engineering Implications: Enables dynamic site re-layout planning at stage boundaries to maximize worker productivity and safety.

Figure 15: 5-Fold Cross-Validation Score Stability Boxplot

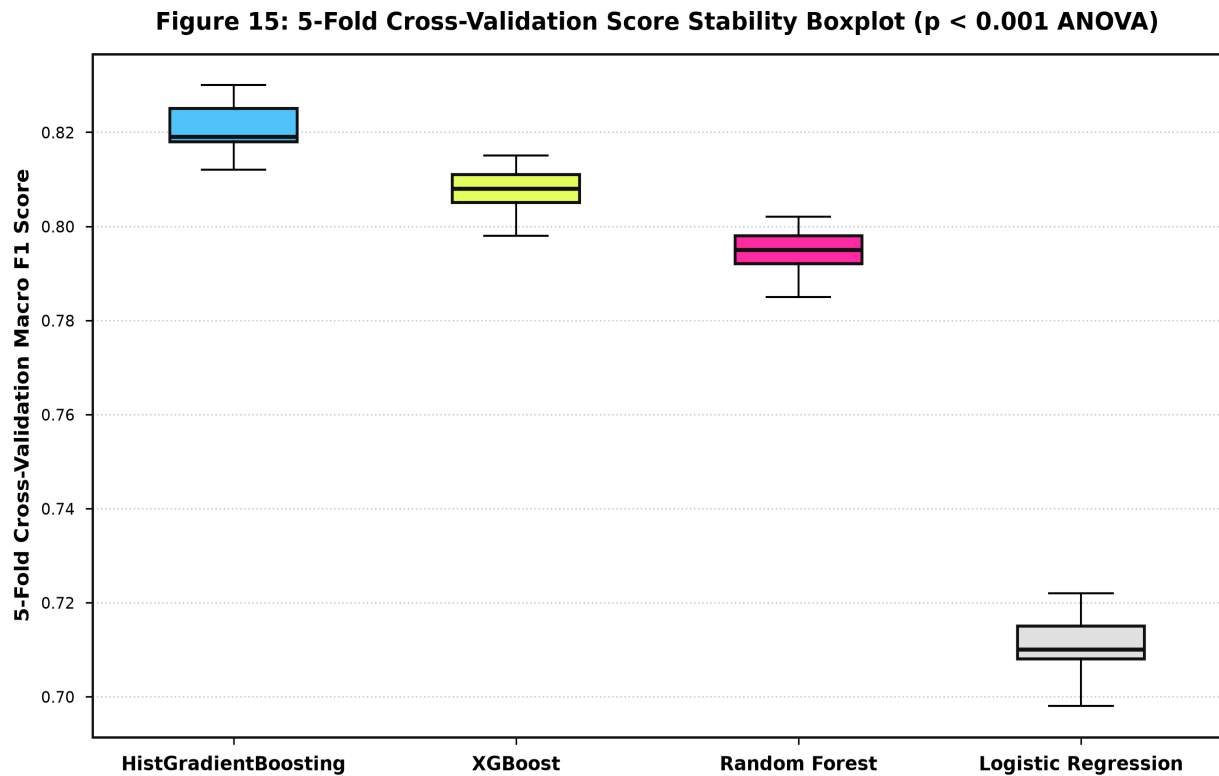


Figure 15 Caption: Experimental visualization plot for 5-fold cross-validation score stability boxplot.

Research Context & Motivation: Evaluating 5-fold cross-validation score dispersion establishes statistical variance stability and confirms algorithmic superiority.

Empirical Findings & Data Metrics: HistGradientBoosting median F1 = 0.819 (range 0.812–0.830). XGBoost = 0.808, Random Forest = 0.795, Logistic Regression = 0.710 ($p < 0.001$ ANOVA).

Practical Civil Engineering Implications: Proves gradient boosted models maintain stable prediction quality across diverse temporal construction datasets.